Self-Learning: ARM Processor Architecture & Features MPCA Module 6

> **Definition:** The **ARM (Advanced RISC Machines)** processor is a 32/64-bit **Reduced Instruction Set Computer (RISC)** architecture engineered for high energy efficiency, low power consumption, and high performance in mobile, IoT, and embedded devices.

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1. Detailed Technical Explanation

1. Key Architectural Features of ARM:

1. **Load-Store Architecture:** Data processing instructions (ADD, SUB, AND) operate **only on registers**; memory is accessed exclusively via explicit `LDR` (Load) and `STR` (Store) instructions.

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2. Register Organization of ARM (ARM7 / Cortex-A)

- Total of **37 registers** (31 General-Purpose 32-bit registers and 6 Status registers), organized into banked registers across operating modes:

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3. CISC (x86/Pentium) vs RISC (ARM) Comparison

| Feature | Intel x86 / Pentium (CISC) | ARM Processor (RISC) |

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4. Quick Recall Flow

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